

AJ C PIPATTANAKUN

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EDUCATION

University of Massachusetts Lowell
Master of Science in Computer Science

Expected Spring 2028
Lowell, MA

University of Massachusetts Lowell
Bachelor of Science in Computer Science, Minor in Data Science

August 2024
Lowell, MA

SKILLS

- **Programming Languages** : C++, C, Java, Python, SQL, JavaScript
- **Software Development** : FastAPI, Pydantic, Node.js, REST APIs, JSON, TCP/IP, object-oriented programming
- **Testing & Validation** : pytest, Google Test, JUnit, requirement-based testing, regression testing, API validation
- **Systems & Tools** : Git, GitHub, Docker, Linux/UNIX, Windows, CMake, Gradle, GitHub Actions, VS Code
- **Collaboration** : Teamwork, Task ownership, pull requests, team updates, documentation review, communicating blockers, and supporting team workflow
- **Communication** : Clear technical notes, progress updates, and team discussions

RELEVANT EXPERIENCE

Software Engineering Intern
Tihranix

May 2026 – November 2026
Foxborough, MA

- Added Python schemas and FastAPI service logic for scenario previews, including ontology references, risk levels, approval flags, and recommended actions.
- Wrote pytest tests to verify scenario behavior, expected API responses, and requirement-based validation for operational intelligence features.
- Validated API endpoints and JSON responses, debugged local runtime issues, and submitted changes through Git feature branches and GitHub pull requests.
- Reviewed documentation and provided team updates on completed tasks, blockers, technical progress, and next steps.

Web Developer
Bang Na Thai Kitchen

May 2022 – June 2025
North Billerica, MA

- Built and maintained the restaurant website using HTML, CSS, and JavaScript; updated menu and contact content and tested the website on desktop and mobile devices.

PROJECTS

Real-Time Multi-Sensor Monitoring and Fault-Tolerant System
C++, Python, Linux, TCP/IP, CMake, Google Test

January 2026 – February 2026

- Developed a multithreaded C++ application that receives simulated temperature, voltage, position, and motion data over TCP/IP and passes messages through thread-safe queues.
- Added fault detection, structured logging, state transitions, and a Python fault-injection tool; wrote Google Test unit, integration, regression, and requirement-based tests.

Secure Command-and-Control Message Processing Simulator
Java, Python, Linux, TCP/IP, JUnit, Gradle, Docker

March 2026 – April 2026

- Built a Java-based simulator that sends structured command messages between a control station and a simulated remote unit over TCP/IP.
- Added message validation, acknowledgements, timeouts, retries, duplicate-message checks, sequence-number checks, audit logs, and JUnit tests for expected system behavior.

Sensor Data Logging and Test Platform
Python, FastAPI, PostgreSQL, Docker, GitHub Actions

May 2026 – June 2026

- Built a test-data logging platform to collect, store, and review simulated sensor and system test results across repeated runs.
- Added API endpoints, database storage, Docker setup, and basic CI checks to support repeatable test review, issue tracking, and technical documentation.

CERTIFICATIONS & TRAINING

- **UMass Global Data Science Bootcamp** — Python, SQL, machine learning, and data visualization August 2024
- **Google IT Support Certificate** — Troubleshooting, networking, operating systems, and IT security January 2021